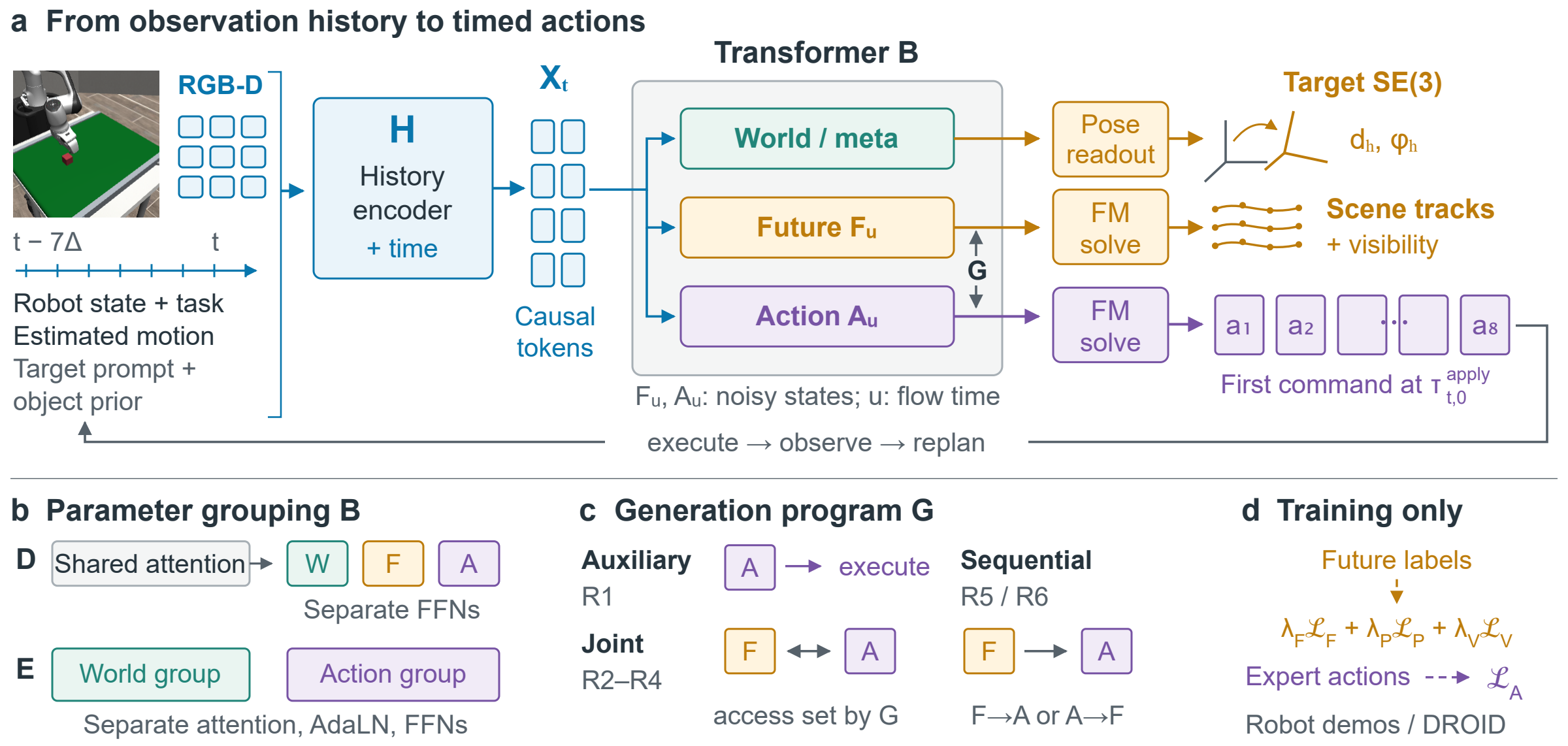

a From observation history to timed actions
RGB-D
t − 7Δ
t
Robot state + task
Estimated motion
Target prompt +
object prior
H
History
encoder
+ time
X_t
Causal
tokens
Transformer B
World / meta
Future F_u
Action A_u
G
Pose
readout
FM
solve
FM
solve
Target SE(3)
d_h, φ_h
Scene tracks
+ visibility
a_1 a_2 ⋯ a_8
First command at $\tau^{apply}_{t,0}$
F_u, A_u: noisy states; u: flow time
execute → observe → replan
b Parameter grouping B
D
Shared attention
W
F
A
Separate FFNs
E
World group
Action group
Separate attention, AdaLN, FFNs
c Generation program G
Auxiliary
R1
A
execute
Joint
R2–R4
F
A
access set by G
Sequential
R5 / R6
F
A
F→A or A→F
d Training only
Future labels
$\lambda_F\mathcal{L}_F + \lambda_P\mathcal{L}_P + \lambda_V\mathcal{L}_V$
Expert actions
$\mathcal{L}_A$
Robot demos / DROID